

PRCP-1027

SKIN DISORDER PREDICTION

Data Science / Machine Learning Project Documentation

Project Property	Details
Project ID	PRCP-1027
Project	Skin Disorder Prediction
Dataset	dataset_35_dermatology (1).csv
Problem Type	Supervised Multi-class Classification
Target	class - six skin disorder classes
Dataset Size	366 rows x 35 columns (34 predictors + class)
Domain	Healthcare
Primary Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Visual Studio Code, Jupyter Notebook, Streamlit
Team ID	PTID-CDS-SEP-26-11163
Team Members	Keerthana S Prasanna Nazre Fatima Purushotham J

1. Executive Summary

This project focuses on the analysis and machine-learning prediction of erythemato-squamous skin diseases. The project brief describes a multi-class healthcare classification problem involving six clinically similar diseases: psoriasis, seborrheic dermatitis, lichen planus, pityriasis rosea, chronic dermatitis, and pityriasis rubra pilaris.

The supplied implementation files provide a working data-loading, preprocessing, train-test split and Random Forest prediction workflow. The supplied CSV contains 366 rows and 35 columns: 34 predictor attributes and the class target.

2. Problem Statement

Task 1: Prepare a complete data analysis report on the given data.

Task 2: Create a predictive model using machine-learning techniques to predict the various classes of skin disease.

Task 3: Provide suggestions to doctors to identify the skin diseases of a patient at the earliest.

3. Project Objectives

- Understand the clinical and histopathological attributes in the dataset.
- Perform exploratory data analysis and identify useful patterns.
- Prepare the data for multi-class classification.
- Implement and compare multiple machine-learning classifiers.
- Evaluate model performance using appropriate classification metrics.
- Document modelling challenges and the techniques used to address them.
- Provide decision-support suggestions for earlier identification without replacing clinical diagnosis.

4. Dataset Overview

Property	Value
Rows in supplied CSV	366
Columns in supplied CSV	35
Predictor attributes	34
Target attribute	class
Target classes	1 to 6
Age field	Linear / numeric after preprocessing
Other feature scale	0 to 3
Family history	0 or 1

4.1. Dataset Sample

The table below shows a sample of the first 8 records from the supplied CSV file. Representative clinical fields, Age, class and the mapped disease name are shown.

erythema	scaling	itching	family_history	Age	class	Skin Disorder
2	2	3	0	55	2	Seborrheic Dermatitis
3	3	2	1	8	1	Psoriasis
2	1	3	0	26	3	Lichen Planus
2	2	0	0	40	1	Psoriasis
2	3	2	0	45	3	Lichen Planus
2	3	0	0	41	2	Seborrheic Dermatitis
2	1	2	0	18	5	Chronic Dermatitis
2	2	3	0	57	3	Lichen Planus

5. Domain and Dataset Context

The project brief explains that erythemato-squamous diseases are difficult to differentiate because they share clinical features such as erythema and scaling, with relatively small differences between conditions. A biopsy is often necessary, but histopathological features can also overlap.

Patients are first evaluated clinically using 12 features. Skin samples are then evaluated using 22 histopathological features under a microscope. The project brief notes that a disease can resemble another disease during an early stage and later develop more characteristic features.

The six disease classes listed in the brief are psoriasis, seboric dermatitis, lichen planus, pityriasis rosea, cronic dermatitis, and pityriasis rubra pilaris.

6. Attribute Information

No.	Clinical Attribute
1	erythema
2	scaling
3	definite borders
4	itching
5	koebner phenomenon
6	polygonal papules
7	follicular papules
8	oral mucosal involvement
9	knee and elbow involvement
10	scalp involvement
11	family history (0 or 1)
34	Age (linear)

6.1. Histopathological Attributes

No.	Histopathological Attribute
12	melanin incontinence
13	eosinophils in the infiltrate
14	PNL infiltrate
15	fibrosis of the papillary dermis
16	exocytosis
17	acanthosis
18	hyperkeratosis
19	parakeratosis
20	clubbing of the rete ridges
21	elongation of the rete ridges
22	thinning of the suprapapillary epidermis
23	spongiform pustule
24	munro microabcess
25	focal hypergranulosis
26	disappearance of the granular layer
27	vacuolisation and damage of basal layer
28	spongiosis
29	saw-tooth appearance of retes
30	follicular horn plug
31	perifollicular parakeratosis
32	inflammatory monoluclear infiltrate
33	band-like infiltrate

7. Task 1 - Data Analysis Plan

The project should inspect data shape, data types, unique values, missing values, class balance and distributions before modelling. The supplied CSV has 366 rows and 35 columns, while the application and notebook explicitly preprocess Age and use the class field as the target.

- Inspect dataset shape, data types and unique values.
- Check missing values and invalid values.
- Review class distribution across all six skin-disorder categories.
- Analyze clinical and histopathological feature distributions.
- Inspect Age as a continuous feature and Family History as a binary feature.
- Use correlation/association analysis as an exploratory tool.
- Use boxplots and distributions to identify unusual values.

7.1. Dataset Loading and Preprocessing Code Snippet

The following excerpt is taken from the supplied **app.py** implementation and shows dataset loading, Age handling, target separation and stratified train-test splitting.

```
# Load dataset
data = pd.read_csv("dataset_35_dermatology (1).csv")

# Data preprocessing
data["Age"] = data["Age"].replace("?", np.nan)
data["Age"] = pd.to_numeric(data["Age"], errors="coerce")
data["Age"] = data["Age"].fillna(data["Age"].median())

# Separate features and target
X = data.drop("class", axis=1)
y = data["class"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# Random Forest model
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
```

8. Data Preprocessing and Model Design

- Age values of '?' are replaced with missing values, converted to numeric and imputed using the median.
- The class column is separated as the target and the remaining columns are used as predictors.
- An 80/20 train-test split is used with random_state = 42 and stratify = y.
- StandardScaler is used in the notebook for Logistic Regression, SVM and KNN.
- Tree-based models such as Decision Tree and Random Forest use the unscaled features.

8.1. Random Forest Implementation Snippet

```
model = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42  
)  
  
model.fit(X_train, y_train)  
  
y_pred = model.predict(X_test)  
accuracy = accuracy_score(y_test, y_pred)  
  
print(f"Model Accuracy: {accuracy:.2%}")
```

9. Candidate Machine-Learning Models

Model	Reason for inclusion
Logistic Regression	Interpretable baseline for multi-class classification
Decision Tree	Captures nonlinear relationships and is easy to explain
Random Forest	Ensemble model that can capture nonlinear interactions and provide feature importance
K-Nearest Neighbors (KNN)	Distance-based classifier for comparison
Support Vector Machine (SVM)	Margin-based classifier for scaled numeric features

10. Evaluation Metrics

Metric	Purpose
Accuracy	Overall proportion of correctly classified observations.
Precision	Correctness of positive predictions for a class.
Recall / Sensitivity	Proportion of actual class observations correctly detected.
F1 Score	Harmonic mean of precision and recall.
Macro F1	Gives equal weight to all disease classes.
Confusion Matrix	Shows class-level correct and incorrect predictions.

10.1. Model Performance Comparison

The supplied notebook creates a model-comparison table and accuracy comparison chart. The actual performance values should be taken from the executed notebook output. The supplied source files do not contain a final completed metrics table, so no unsupported metric values are inserted here.

11. Confusion Matrix Analysis

For this six-class problem, confusion matrices should be reviewed for every classifier. They show which disease classes are confused with each other, especially where clinical and histopathological characteristics overlap.

- Review class-wise true positives and false negatives.
- Identify disease pairs that are frequently confused.
- Use class-wise recall to identify difficult categories.
- Use macro-averaged metrics so smaller classes are not hidden by overall accuracy.

12. Feature Importance / Model Explainability

Random Forest feature importance can be used to identify attributes that contribute most strongly to the model's predictions. The actual ranking should be reported from the executed notebook.

13. Task 3 - Suggestions to Doctors for Earlier Identification

The project brief emphasizes that visually similar skin diseases may be difficult to differentiate and that early-stage disease may resemble another condition. Suggestions are therefore framed as decision-support guidance.

- Consider the complete set of clinical characteristics rather than relying on a single symptom.
- Include family history where available.
- Consider age together with the clinical and histopathological features.
- When clinical features are ambiguous, correlate them with histopathological findings.
- Use the machine-learning output as an additional decision-support signal and confirm diagnosis through appropriate clinical evaluation.

14. Report on Challenges Faced

Challenge	Recommended technique / reason
Overlapping clinical symptoms	Use multiple features rather than single-symptom rules.
Histopathological overlap	Combine clinical and histopathological evidence where available.
Early-stage similarity	Use multi-class classification and class-level metrics.
Different feature scales	Scale numeric inputs for KNN, SVM and linear models.
Multi-class target	Use stratified splitting and macro-averaged evaluation.
Model instability	Use cross-validation and confusion-matrix analysis before production use.

15. Deployment / Streamlit Application

The supplied app.py creates a Streamlit application titled 'Skin Disorder Detection'. It loads the CSV, preprocesses Age, trains a Random Forest classifier with 100 trees, displays model accuracy, accepts patient feature inputs and returns a predicted skin-disorder class.

16. Technologies Used

Technology	Purpose
Python	Core programming language
Pandas	Data loading, cleaning and transformation
NumPy	Numerical computation
Matplotlib / Seaborn	Exploratory visualization
Scikit-learn	Preprocessing, classifiers, metrics and model development
Visual Studio Code	Code editing and development
Jupyter Notebook	Interactive analysis and model development
Streamlit	Interactive prediction application

17. Application Screenshots

The following screenshots document the implemented Streamlit Skin Disorder Detection application.

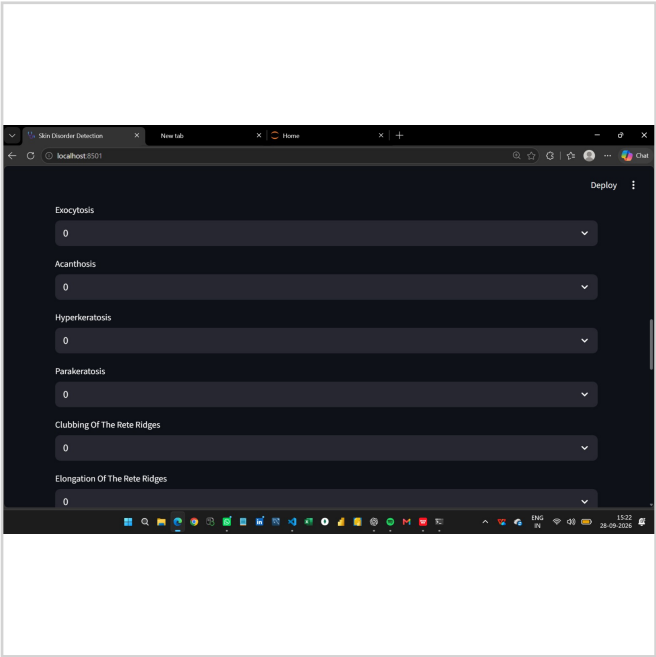


Figure 17.1 - Application Home / Accuracy

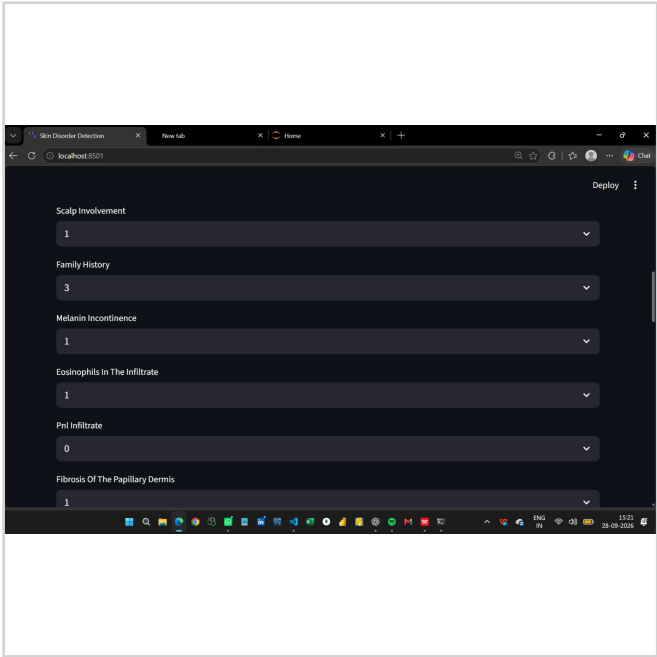


Figure 17.2 - Patient Input - Clinical Features

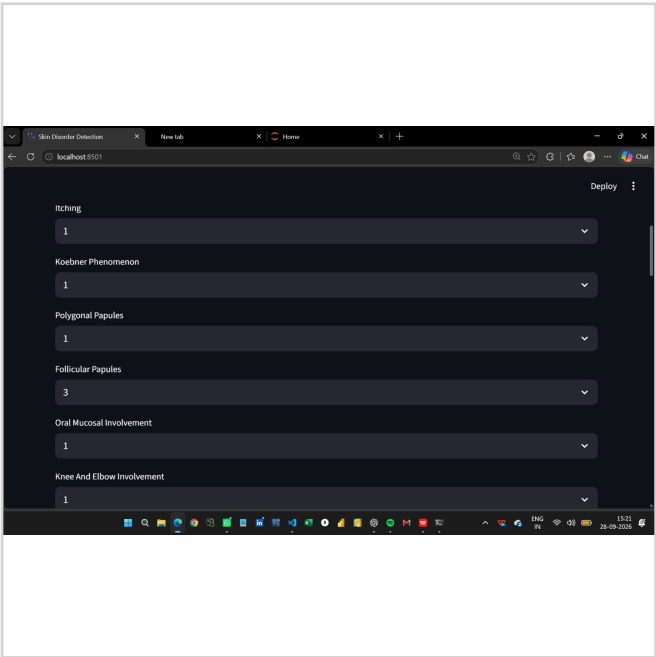


Figure 17.3 - Patient Input - Additional Features

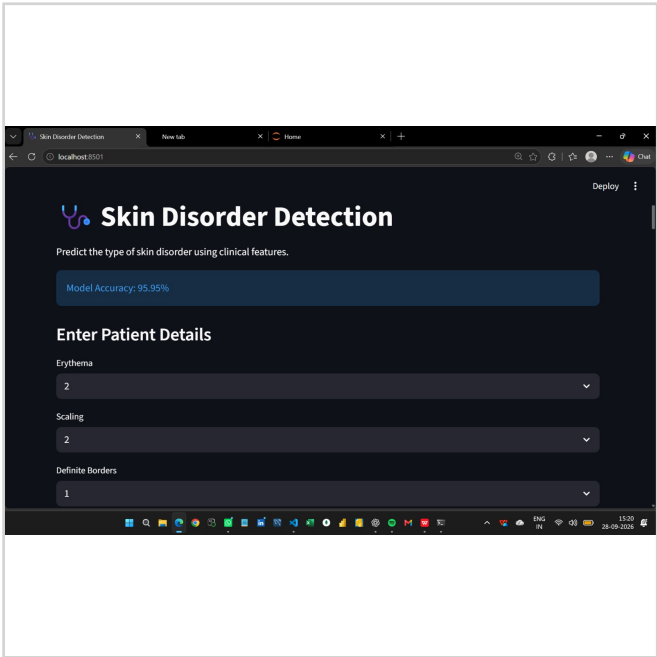


Figure 17.4 - Patient Input - Histopathological Features

17. Application Screenshots - Continued

Additional screenshot of the Streamlit prediction result.

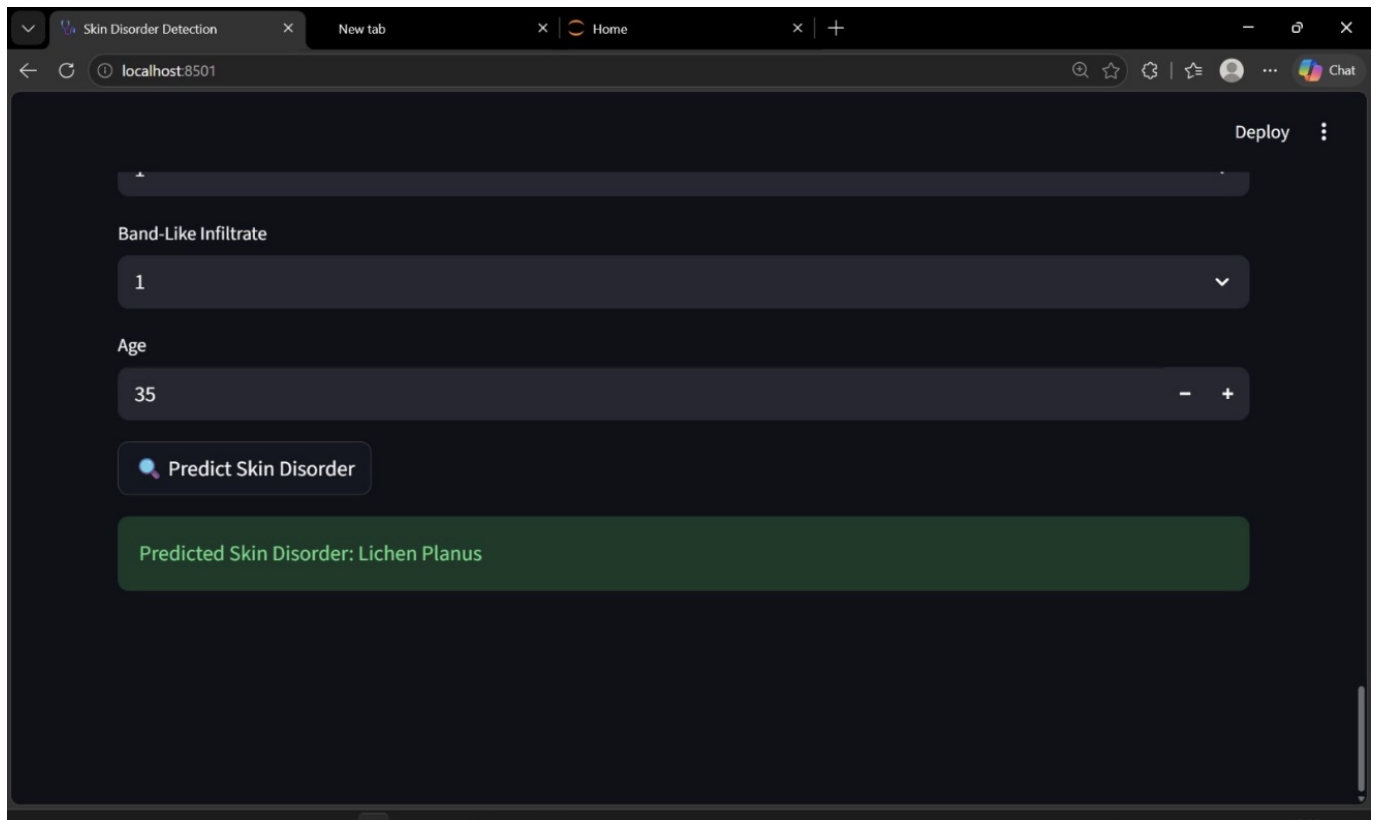


Figure 17.5 - Streamlit prediction result

18. Project Run / Submission Guidance

- Install Python 3.x and the required data-science libraries.
- Place dataset_35_dermatology (1).csv in the project directory.
- Open the single Jupyter notebook used for final submission.
- Run data loading, EDA, preprocessing, training, model comparison and evaluation cells from top to bottom.
- Confirm that charts, confusion matrices and model-performance outputs are populated before submission.
- Run the Streamlit application using the supplied app.py when the interactive prediction page is required.

19. Limitations

- The supplied source describes clinically similar diseases with overlapping features.
- The dataset contains multiple classes and may not have equal class sizes.
- The model depends on the quality and availability of the supplied clinical and histopathological attributes.
- The prediction should not be considered a replacement for professional medical diagnosis.
- External validation is required before real-world clinical deployment.

20. Conclusion

PRCP-1027 defines a multi-class healthcare machine-learning problem for distinguishing six erythemato-squamous skin diseases using 34 predictor attributes and a class target. The supplied implementation demonstrates data loading, Age preprocessing, stratified splitting and Random Forest prediction through a Streamlit interface.

The supplied CSV provides 366 records and 35 columns. The report includes a real dataset sample and representative Python implementation snippets from the supplied project files so that the documentation reflects the implemented workflow.

Model-performance values should be populated from the fully executed notebook before final submission. The project should use the completed class-level evaluation, confusion matrices and validation results to support any final production recommendation.

21. Source Project Information

Project: PRCP-1027 - Skin Disorder

Domain: Healthcare

Implementation files supplied: app.py, dataset_35_dermatology (1).csv, Untitled (1).ipynb.